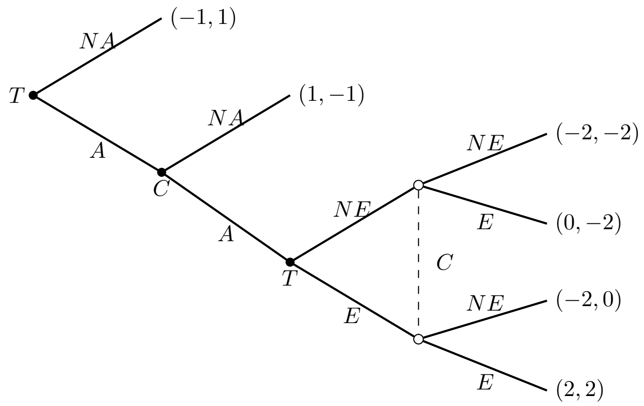
Subgame Perfect Nash Equilibrium

Consider the following game between Trom T and Chi C. First, T decides whether to raise tariffs, A, or not raise tariffs, NA. If T raises tariffs, then C observes this action and decides whether to also raise tariffs, A, or not raise tariffs, NA. If C also raises tariffs, both players enter a simultaneous negotiation round in which each decides whether to eliminate all tariffs, E, or not eliminate them, NE. Payoffs are ordered as (T, C).



1. Write the actions and strategies of each player
2. Find the subgame-perfect Nash equilibrium or equilibria
3. Write the game in strategic form and find the Nash equilibria in pure strategies. What difference do you find with respect to part b)? Why does this happen?

Solution

1. Player T has two decision nodes. First, T chooses whether to raise tariffs:

$$A \qquad NA$$

If the negotiation stage is reached, T chooses:

$$E \qquad NE$$

Therefore, T's pure strategies are complete contingent plans:

$$S_T = \{(NA, NE), (NA, E), (A, NE), (A, E)\}$$

Player C also has two decision moments. After observing that T raised tariffs, C chooses:

$$A \qquad NA$$

If the negotiation stage is reached, C chooses:

$$E \qquad NE$$

Therefore, C's pure strategies are:

$$S_C = \{(NA, NE), (NA, E), (A, NE), (A, E)\}$$

Conclusion: each player's strategy must specify what to do at every decision point, including decision points that may not be reached in equilibrium

2. We solve by backward induction. If both countries raised tariffs, the final simultaneous negotiation game is:

	NE_C	E_C
NE_T	$(\underline{-2}, \underline{-2})$	$(0, \underline{-2})$
E_T	$(\underline{-2}, 0)$	$(\underline{2}, \underline{2})$

This subgame has two pure-strategy Nash equilibria:

$$(NE, NE) \qquad (E, E)$$

First, suppose the continuation equilibrium is (NE, NE). Then, if C chooses A, the continuation payoff is

$$(-2, -2)$$

At C's first decision node, C compares:

$$NA: \; -1 \qquad A: \; -2$$

Therefore, C chooses NA. Then T compares:

$$NA: \; -1 \qquad A: \; 1$$

Therefore, T chooses A.

So one subgame-perfect Nash equilibrium is:

$$s_T = (A, NE) \qquad s_C = (NA, NE)$$

Now suppose the continuation equilibrium is (E, E). Then, if C chooses A, the continuation payoff is

$$(2, 2)$$

At C's first decision node, C compares:

$$NA: \ -1 \qquad A: \ 2$$

Therefore, C chooses A. Then T compares:

$$NA: \ -1 \qquad A: \ 2$$

Therefore, T chooses A.

So another subgame-perfect Nash equilibrium is:

$$s_T = (A, E) \qquad s_C = (A, E)$$

Conclusion: the game has two subgame-perfect Nash equilibria in pure strategies:

$$s_T = (A, NE) \qquad s_C = (NA, NE)$$

and

$$s_T = (A, E) \qquad s_C = (A, E)$$

3. The strategic form is obtained by listing each player's complete strategies.

	(NA, NE)	(NA, E)	(A, NE)	(A, E)
(NA, NE)	$(-1, \underline{1})$	$(-1, \underline{1})$	$(\underline{-1}, \underline{1})$	$(-1, \underline{1})$
(NA, E)	$(-1, \underline{1})$	$(-1, \underline{1})$	$(\underline{-1}, \underline{1})$	$(-1, \underline{1})$
(A, NE)	$(\underline{1}, \underline{-1})$	$(\underline{1}, \underline{-1})$	$(-2, -2)$	$(0, -2)$
(A, E)	$(\underline{1}, -1)$	$(\underline{1}, -1)$	$(-2, 0)$	$(\underline{2}, \underline{2})$

The Nash equilibria in pure strategies are:

$$((A, NE), (NA, NE))$$

$$((A, NE), (NA, E))$$

$$((NA, NE), (A, NE))$$

$$((NA, E), (A, NE))$$

$$((A, E), (A, E))$$

Among these, the subgame-perfect Nash equilibria are only:

$$((A, NE), (NA, NE)) \qquad ((A, E), (A, E))$$

The difference is that Nash equilibrium in strategic form only requires optimality along the equilibrium path. It can include non-credible behavior in subgames that are not reached. Subgame perfection eliminates those equilibria by requiring Nash equilibrium behavior in every subgame.

Conclusion: there are more Nash equilibria in the strategic form than subgame-perfect Nash equilibria, because some Nash equilibria rely on non-credible off-path behavior. Subgame perfection removes those equilibria